

Assignment #5: Arduino Pins and Their Role in Embedded Systems

Arduino boards provide three main types of pins: Digital Pins, Analog Pins, and PWM-enabled Pins. These pins enable interaction with real-world systems by supporting sensing, decision-making, and control functions.

Digital Pins:

Digital pins operate in binary mode (HIGH or LOW) and are used for simple input/ output tasks such as reading button states or controlling LEDs. They form the foundation for logic-based decision-making in embedded systems.

Analog Pins:

Analog pins read continuous voltage levels using an Analog-to-Digital Converter (ADC). This allows Arduino to process real-world signals such as temperature, light intensity, or sound. Analog inputs are essential for sensing environmental data.

PWM (Pulse Width Modulation) Pins:

PWM pins simulate analog output using digital signals by varying the duty cycle of a square wave. This is useful for controlling devices like motors, LEDs (brightness control), and actuators. PWM bridges the gap between digital control and analog-like behavior.

Collective Role in Embedded and IoT Systems:

- Sensing: Analog pins capture real-world data from sensors.
- Decision-Making: The microcontroller processes input data using programmed logic.
- **Control:** Digital and PWM pins drive outputs such as motors, displays, and communication modules.

In IoT systems, these pins allow Arduino to collect sensor data, process it, and communicate with other devices or cloud platforms. For example, a smart home system may use analog pins to read temperature, digital pins to detect motion, and PWM pins to control lighting intensity.

Conclusion:

Together, digital, analog, and PWM pins provide a complete interface for interacting with the physical world. They enable embedded systems to sense inputs, process information, and produce meaningful outputs, making Arduino a powerful platform for prototyping and real-world applications.

